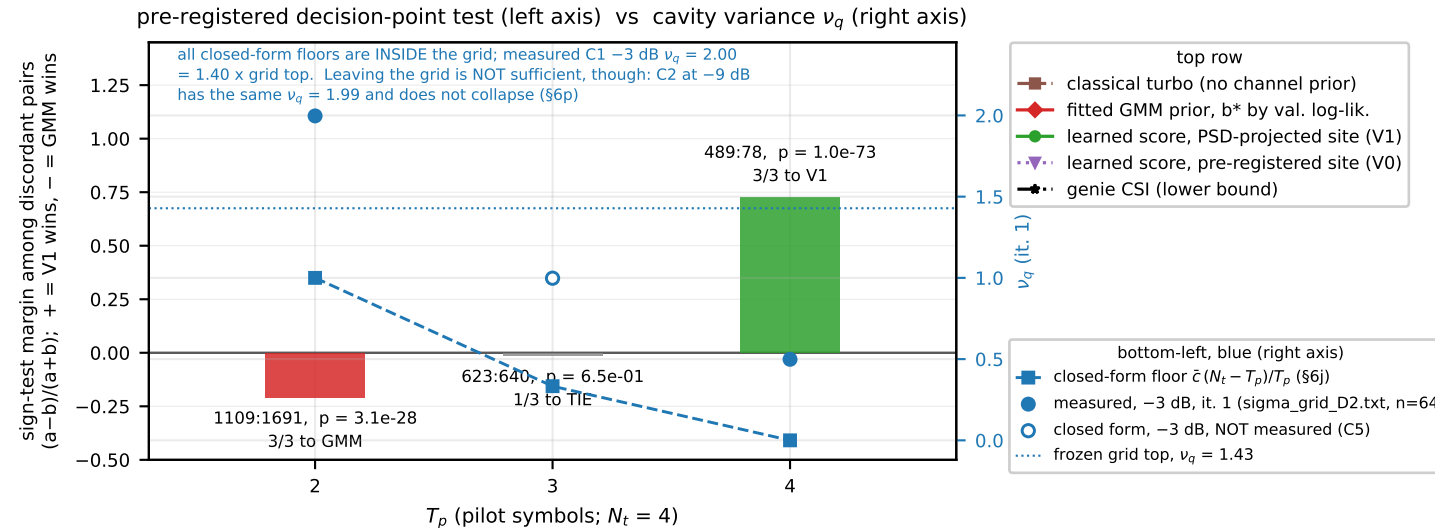
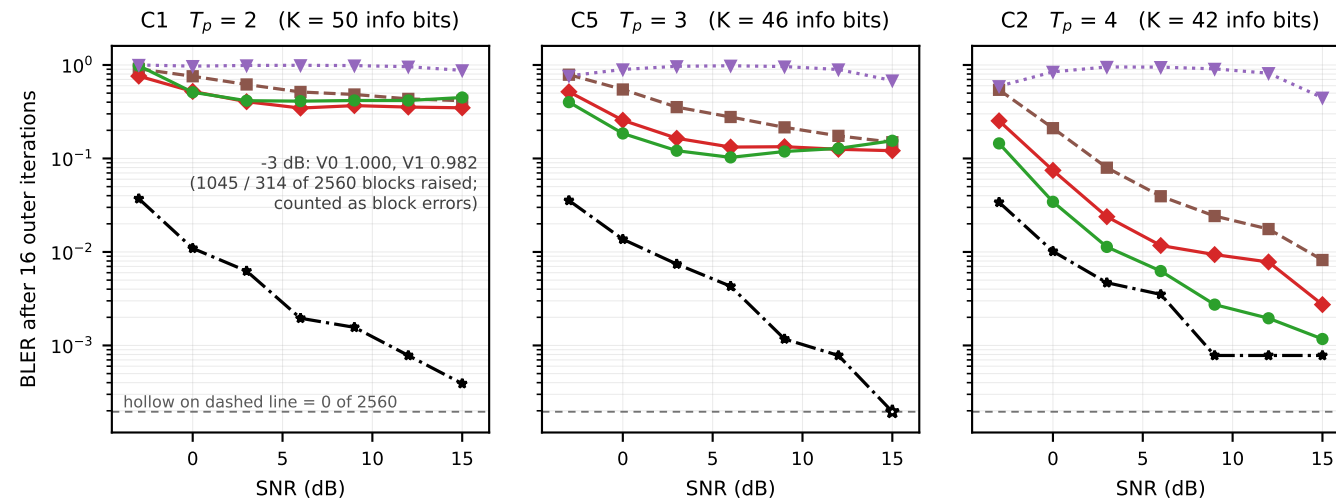




decision SNRs (anchor rule): C1 and C5 at +6/+12/+15 dB, C2 at -3/+0/+3 dB; C5 reverses at +6 dB (232:156, $p = 1.3e-04$)

EQUAL BUDGET: every arm, learned and classical, is fitted / trained on the SAME $N_{\text{train}} = 1.6e5$ channel set (10_SPEC_stageC A3). $n = 2560$ trials per SNR point. GMM b* = kron $K=512$, by validation log-likelihood over all 12 configurations of the $K \leq 512$ grid; the extended $K=1024$ grid moves b* (F15c).

checkpoint ckpt/d2sx_N160000_a1.pt: gates are measurable on D1 only, so this D2 checkpoint is qualified through its D1 sibling sx_N160000_D1.pt, which PASSES GB 2.66e-3 / GC 0.0999 / GD 0.0718 (LADDER_C.md:1).

At this budget C2's GMM improves to 0.2434 on the extended $K=1024$ grid (results/tables_D2_B16e4k.txt); V1 is 0.1449 there. See F15.